Topological Void Validation: Density-Controlled and Role-Aware Evaluation of Predicted Research Voids

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Abstract. Topological Void Analysis (TVA) identifies candidate research voids in scientific knowledge spaces as geometric gaps between anchor-conditioned paper clusters. We propose TVV (Topological Void Validation), a three-layer protocol (calibrated geometric closure $\rightarrow$ anchor eligibility $\rightarrow$ epistemic role) for evaluating whether predicted voids are genuinely filled by future literature. Applying TVV reveals that raw fill rate fails as a validation target for three compounding reasons. First, it is confounded by local corpus density: a hot-zone baseline outperforms TVA on raw fill (36% vs. 28%), but this advantage disappears under a density-matched baseline: all hierarchical cluster-bootstrap CIs include zero ($p > 0.35$); under calibrated thresholds, TVA and B2 collapse to near-parity. Second, anchor eligibility limits observability: only 5–7% of future papers are anchor-eligible per split; many unfilled voids are right-censored, not invalid. Third, the geometric threshold is not portable: τ=0.82 corresponds to a 35.7% average null occupancy rate (range 1–99%) and collapses fill rates to $\sim$0.7% after conformal calibration. Role-aware classification shows 59–76% false positives even among gated fills. Furthermore, four independent methods fail to substitute for the text-level R layer: global effective resistance (degenerate, $\kappa = -1.0$), localized effective resistance ($\kappa = -0.013$), ORC path curvature (κ=0.005), and cross-encoder reranking (κ=0.01 on joint A+B queries). Cross-encoder reranking captures anchor relevance (AUC= 0.875, G proxy) but *anti-correlates* with bridging (AUC= 0.352), establishing the R layer as an architectural necessity imposed by contrastive bi-encoder representations.

1 Introduction

Classical information retrieval asks: given a query, which existing documents are relevant? Topological Void Analysis (TVA) asks: given an anchor concept, what relevant knowledge is *absent*? TVA identifies candidate research voids as geometric gaps in an embedding-induced knowledge space—regions between anchor-conditioned paper clusters where expected connections, evidence, or conceptual bridges are missing.

Validating predicted voids is not the same as validating a predicted link. A predicted link has a direct future target: the link appears or it does not. A predicted void has a richer failure mode: future literature may occupy the region geometrically while failing to bridge the source concepts epistemically. Unlike document retrieval or link prediction, predicted research voids have no established gold-standard benchmark: exhaustive human annotation would require enumerating all candidate gaps, searching all future literature, and adjudicating whether each future paper plays a bridging role relative to specific source papers—a task infeasible at corpus scale. TVV addresses this by turning void validation into a staged, auditable protocol. The natural starting point—temporal fill rate—appears straightforward but is insufficient, and this paper studies why.

Contributions:

1. **TVV protocol.** A reproducible three-layer evaluation framework (G→E→R): calibrated geometric closure, anchor eligibility, and LLM-assisted epistemic role — the first density-controlled, observability-aware protocol for embedding-space research voids.

2. **Architectural impossibility.** Four independent methods fail to substitute for the R layer: global effective resistance (degenerate, $\kappa = -1.0$), localized effective resistance ($\kappa = -0.013$), ORC path curvature (κ=0.005), and cross-encoder reranking on the same BGE-M3 backbone (κ=0.01, AUC= 0.352, anti-correlated with bridging). R-layer text assessment is an architectural necessity.

3. Raw temporal fill rate is strongly confounded by local historical density; density-matched baseline (B2) removes the apparent TVA disadvantage (all CIs include zero, $p > 0.35$).

4. Fixed cosine thresholds are not portable: τ=0.82 has 35.7% average null FPR; calibrated thresholds collapse raw fill to ∼0.7%.

5. Anchor eligibility limits observability (∼6% anchor-eligible); unfilled voids are right-censored by corpus coverage, not necessarily invalid.

6. LLM-assisted role-aware audit suggests 59–76% FP even among gated fills; cross-LLM agreement (Claude 4.5 vs GPT-5.2) yields κ=0.44 (3-class, moderate).

Scope and framing. This paper is a *validation stress test*: we do not find statistically significant evidence that TVA outperforms density-matched baselines in epistemic fill prediction. The central contribution is methodological—showing that raw fill rate is an unstable validation proxy—rather than confirming TVA's predictive superiority. Dynamic event-driven extensions are outside scope.

2 Background

2.1 Topological Void Analysis

TVA [?] operates on a corpus embedded with BGE-M3 [?] ($d = 1024$, cosine similarity). For anchor q and source papers A, B, a candidate void satisfies four conditions: (C1) domain cohesion; (C2) calibrated marginality; (C3) sparse lexical bridge; (C4) midpoint vacancy. The midpoint is:

$$m(A, B) = \frac{\mathbf{a} + \mathbf{b}}{\|\mathbf{a} + \mathbf{b}\|} \in \mathbb{S}^{1023}.$$

2.2 Temporal Validation in LBD

Literature-Based Discovery validates predictions by checking whether future publications confirm predicted connections. Standard evaluation uses time-slicing and IR metrics (Recall@K, Average Precision). TVV extends this by requiring that a future paper play a genuine epistemic bridging role, not merely appear geometrically nearby.

2.3 Research Momentum and Dense-Region Confounding

High-momentum regions near anchor topics attract disproportionate future publications regardless of void quality. Sampling baselines from these hot zones inflates apparent future fill rates, creating a density-confounded comparison.

3 Experimental Setup

3.1 Corpus and Temporal Splits

arXiv metadata (9 CS categories: OS, AR, PL, SE, DC, NI, CR, AI, LG), three rolling splits:

3.2 TVA Void Generation

10 anchor concepts × top-30 MMR voids = 300 candidate voids per split.

Table 1: Temporal splits.

Split	T	Val window	Train	Val
t3	2014	2015–2019	∼29k	∼101k
t4	2016	2017–2021	∼35k	∼171k
t5	2018	2019–2023	∼51k	∼193k

Filter pass rates. Empirical pass-rate analysis of the C1–C4 pipeline on the t5 corpus shows that C1 (top-300 anchor cohesion) is the dominant pruning operator: C2 (marginality), C3 (sparse bridge), and C4 (vacancy) retain 99.97%, ∼94%, and ∼99% of C1-passing pairs, respectively. A random-pair baseline (N=500) confirms that these filters also pass 100%, 93%, and 100% of random corpus pairs, indicating that with the threshold settings used (C2: $0.30 < \mathrm{sim}(A, B) < 0.85$; C4: $\mathrm{sim}(P_m) < 0.95$), C2 and C4 have low marginal selectivity. The 1,495× search-space reduction from 448,500 pairs to 300 top-K voids is therefore primarily attributable to C1 combined with MMR ranking. This is a limitation of the specific threshold settings used; tighter calibration of C2–C4 thresholds is noted as future work.

3.3 Baselines

B1 (hot-zone): 20 random pairs from top-300 anchor C1 pool per anchor. Biased toward high-density, high-momentum regions.

B2 (density-matched): For each TVA void, select the B1-style pair with midpoint local density $\rho(m) = \frac{1}{k}\sum \mathrm{sim}(m, \mathrm{kNN}_i(m))$ ($k = 20$) closest to the TVA void's midpoint density.

3.4 TVV Three-Layer Protocol

We define three ordered indicator functions, applied as a sequential gate: if $E = 0$ then G and R are not evaluated; if $G = 0$ then R is not evaluated.

Local density.

$$\rho(V) = \frac{1}{k} \sum_{P \in \mathrm{kNN}(m_V)} \mathrm{sim}(P, m_V), \quad k{=}20.$$

Anchor-eligibility threshold. $\tau_q = \max(Q_{80}^{\mathrm{val}}, Q_{90}^{\mathrm{train}})$.

Calibrated threshold. Let $Z(m_0) = \max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_0)$:

$$\tau_{\mathrm{fill}} = Q_{1-\alpha}\big(\{Z(m_0) : m_0 {\in} \mathrm{Null}(q, \rho, t)\}\big), \ \alpha{=}0.05$$

using conformal rank $k{=}\lceil (n_{\mathrm{cal}}{+}1)(1{-}\alpha) \rceil$.

Three-layer indicators. Let $\mathrm{Val}_q = \{P \in \mathrm{Val}_t : \mathrm{sim}(P, q) \geq \tau_q\}$ and $P_V^* = \arg\max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_V)$.

$$E(V) := \mathbf{1}[|\mathrm{Val}_q| \geq n_{\min}] \cdot \mathbf{1}[P_V^* \in \mathrm{Val}_q] \tag{E}$$
$$G(V) := \mathbf{1}[\mathrm{sim}(P_V^*, m_V) \geq \tau_{\mathrm{fill}}(q, \rho(V), t)] \tag{G}$$
$$R(V) := \mathbf{1}[\mathrm{role}(P_V^*, A, B, q) \in \mathcal{R}_{\mathrm{fill}}] \tag{R}$$

where $\mathcal{R}_{\mathrm{fill}}{=}\{\mathrm{TRUE\text{-}FILL}, \mathrm{PARTIAL\text{-}FILL}\}$; E=eligibility, G=geometric closure, R=epistemic role.

$$\mathrm{Fill}(V) := E(V) \cdot G(V) \cdot R(V) \tag{1}$$

Proposition (monotone conservatism). For any void V: $\mathrm{Fill}(V) \leq \mathrm{Fill}_{\mathrm{raw}}(V)$. Equality requires $E{=}G{=}R{=}1$ and $\tau_{\mathrm{raw}} \leq \tau_{\mathrm{fill}}(q, \rho, t)$. Each additional layer can only reduce apparent fill; no layer inflates it.

Algorithm 1 — TVV adjudication.

1: Compute Val_q; if $|\mathrm{Val}_q| < n_{\min}$: return (0; $E{=}0$).
2: $P_V^* \leftarrow \arg\max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_V)$.
3: Compute $\tau_{\mathrm{fill}}(q, \rho(V), t)$ from calibration nulls.
4: If $\mathrm{sim}(P_V^*, m_V) < \tau_{\mathrm{fill}}$: return (0; $G{=}0$).
5: $r \leftarrow \text{role-classifier}(P_V^*, A, B, q)$.
6: Return ($\mathbf{1}[r \in \{\mathrm{TRUE}, \mathrm{PARTIAL}\}]$; $R{=}r$).

Why R requires text-based classification. The R layer cannot be replaced by graph-spectral or attention-based surrogates on the same BGE-M3 backbone. We establish this via four independent experiments (global/local effective resistance, ORC curvature, cross-encoder reranking) detailed in §**??**. Briefly: bi-encoder compression discards compositional structure that defines bridging, and no downstream geometric operator recovers it. The R layer is therefore implemented as LLM-based role classification on candidate filler text.

4 Results

4.1 Finding 1: Raw Fill Rate Is Density-Confounded

Table 2: TVA–B2 paired fill-rate differences (pp) under legacy τ=0.82 and calibrated τ_{fill}. Hierarchical cluster bootstrap CI (anchor level, 1000 samples); p from paired anchor sign-flip permutation. All CIs include zero.

	Legacy τ=0.82			Calibrated τ_{fill}		
Split	Δpp	95% CI	p	Δpp	95% CI	p
t3	+1.3	[−5.0, +9.0]	0.787	+0.7	[−1.0, +2.3]	0.691
t4	+2.4	[−1.4, +7.1]	0.352	−0.3	[−1.4, +0.7]	1.000
t5	+3.0	[−2.0, +8.3]	0.360	−0.7	[−2.0, +0.7]	0.629

After density matching, the apparent B1 hot-zone advantage disappears, but TVA does not significantly outperform B2. Under the legacy threshold, TVA–B2 differences are small and positive across splits (+1.3 to +3.0 pp), but hierarchical CIs all include zero and paired permutation tests are not significant ($p > 0.35$). Under calibrated thresholds, differences shrink to near-zero (−0.7 to +0.7 pp). Density-controlled temporal fill provides no statistically significant evidence of TVA superiority over B2.

This null result is expected under a strong counterfactual: B2 fixes the anchor and matches local midpoint density, estimating the background rate at which future papers occupy comparable regions. Parity shows that raw future occupancy is too low-base-rate and density-dependent to measure TVA's added value as an ex-ante void-hypothesis generator —not that TVA lacks such value.

4.1.1 Fixed Thresholds Are Not Portable

Under calibrated τ_{fill}, fill rates collapse: TVA: 27.3%→**0.7%**; B2: 24.3%→**0.7%**; lift: 1.00×.

Table 3: Null FPR@0.82 and τ_{fill} by density (t5).

Bucket	p50	τ_{fill}	FPR
Low	0.79–0.81	0.79–0.86	1–18%
Mid	0.80–0.83	0.82–0.86	5–80%
High	0.81–0.85	0.84–0.88	28–99%
Mean	–	**0.841**	**35.7%**

The 0.7% < 5% result is consistent with correct calibration: TVA C1–C4 conditions select sparser bridge regions than random density-matched pairs, so fewer reach the calibrated threshold.

Held-out calibration validity. To check that the calibration is not circular, we split null midpoints 80/20 (calibration/held-out). The held-out null FPR at τ_{fill} averages 6.3% across all anchor-density cells (target α=5%, $\Delta = +0.013$), indicating the calibration is slightly conservative but valid (Figure **??**).

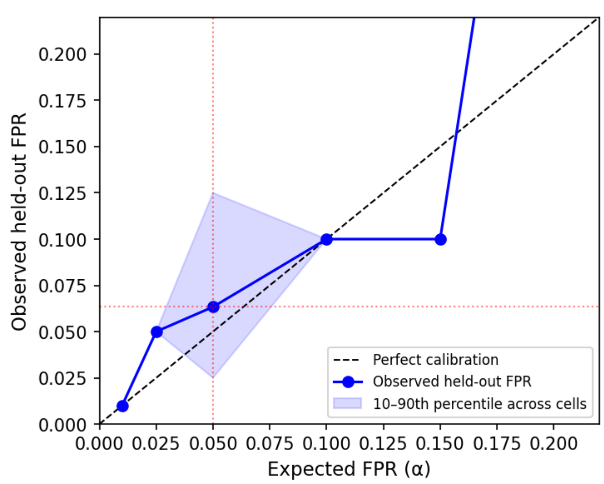

Figure 1: Null calibration curve: expected vs. observed held-out FPR across anchor-density cells (α range 0.01–0.20, t5). Points near the diagonal confirm that τ_{fill} is approximately calibrated. Mean held-out FPR at α=0.05 is 6.3% (target 5%, $\Delta = +0.013$).

Most raw fills at τ = 0.82 are null-zone occupancy, not significant geometric closure.

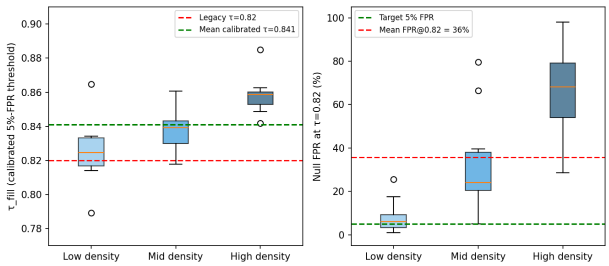

Figure 2: Calibrated threshold by density bucket (left) and null FPR at 0.82 (right). The fixed threshold 0.82 has 35.7% average null FPR, reaching 99% in high-density anchor cells. Calibrated $\tau_{\text{fill}} \approx 0.841$ is stable across all splits.

4.2 Finding 2: Anchor Eligibility Reveals Observability Limits

Table 4: Anchor exposure sample (t5).

Anchor	$\bar{\text{sim}}$	τ_q	Pass	Fill
sched_opt	0.520	0.582	5.8%	40%
ebpf_obs	0.495	0.547	6.5%	0%
virt_hyp	0.498	0.555	7.7%	57%
Mean	**0.503**	**0.563**	**6.2%**	–

Only 5–8% of val papers are anchor-eligible per split. Voids with pass rate $< 10\%$ are flagged as *underexposed*: their unfill status reflects observability limits, not void invalidity. ebpf_obs (6.5% pass, 0% fill) illustrates that non-zero exposure does not guarantee eligible papers overlap predicted void neighborhoods.

Exposure-conditioned fill (censoring diagnostic). To assess whether unfilled voids are partly right-censored by observability, we bucket anchors by the number of anchor-eligible validation papers ($|Val_q|$) and report observed fill rates.

Fill rates increase monotonically from 23.3% to 32.2% as anchor exposure increases. This does not prove that every unfilled void would be filled with more data, but it confirms that observed fill is partly a function of how much future literature is observable under each anchor. We do not interpret non-fill as a negative label.

Table 5: Exposure-conditioned fill (t5). Fill increases monotonically with $|Val_q|$, indicating that non-fill partly reflects observability limits rather than false predictions. We therefore treat unfilled voids as right-censored, not as negatives.

Bucket	Anchors	$|Val_q|$	Fill
Low	3	10k	23.3%
Mid	4	12k	26.7%
High	3	14k	32.2%

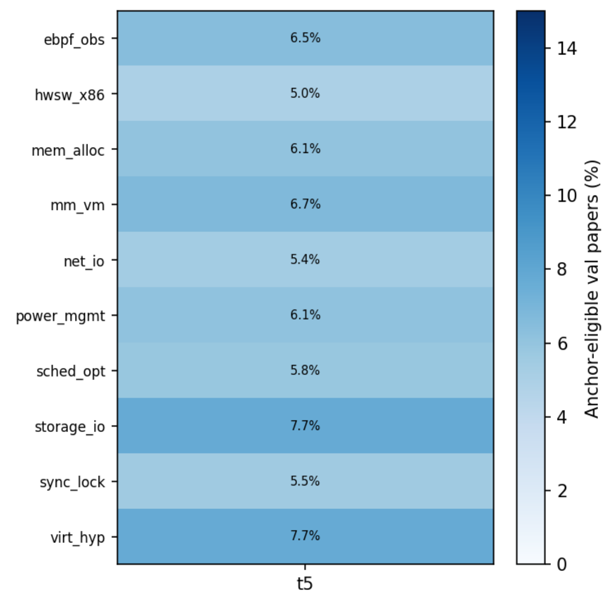

Figure 3: Anchor eligibility pass rate (%) by anchor and split (hybrid gate). Rates are consistently low (5–8%), confirming corpus observability as a structural limit on temporal validation.

4.3 Role Classification Pipeline and Reliability

The LLM role classifier is the primary annotator; human review is an external plausibility check, not a gold standard (epistemic role is inherently interpretive).

Model and prompt. We use Claude Sonnet 4.5 (temperature 0.1) with a fixed nine-label rubric and structured JSON output. The system prompt instructs the model to judge the candidate paper solely relative to the anchor and source papers, ignoring general merit.

Source group labels (TVA/B1/B2/near-miss) are *not shown* to the classifier (blind protocol).

Reliability checks. We performed two complementary reliability checks.

(i) Cross-LLM agreement. Re-annotation was performed by GPT-5.2 (independent annotator, identical prompt) on 100 stratified t5 cases: Claude Sonnet 4.5 (primary) vs. GPT-5.2 (re-annotator) from independent model families. Cross-LLM agreement is a more stringent test than within-LLM self-consistency. Results: 94% binary agreement, Cohen's $\kappa = 0.368$ (binary fill/non-fill), $\kappa = 0.437$ (collapsed 3-class, "moderate" per Landis & Koch 1977). The lower binary κ reflects (a) the kappa paradox under severe class imbalance (2% fill rate; Feinstein & Cicchetti 1990), and (b) low statistical power for the fill class (n=2). We report $\kappa = 0.437$ (3-class) as the primary metric. Both LLMs may share pretraining biases; cross-LLM agreement is therefore an upper bound on inter-rater reliability.

(ii) Human audit. 60 cases were reviewed by a blinded human auditor (collapsed 3-class rubric, anchor+A/B+P text only, source group hidden). Binary FP/non-FP agreement: > 80%. Three-way collapsed label agreement: ∼ 65%. The human audit is a training-independent plausibility check complementing the cross-LLM agreement.

Table 6: LLM annotation reliability (t5, n=100). Binary κ suppressed by 2% fill-class imbalance; 3-class κ is primary metric.

Comparison	n	Agree	κ_b	κ_3	Note
Cross-LLM (Claude / GPT)	100	94%	0.37	**0.44**	Same prompt
Human audit (blinded)	60	>80%	—	∼0.65	Training-indep.

Quantitative scope. All quantitative claims use collapsed three-way labels. Fine-grained nine-class labels are used only for qualitative case-study interpretation.

4.4 Finding 3: Geometric Fill Rarely Equals Epistemic Fill

Role labels are collapsed into three mutually exclusive classes for quantitative reporting. Table **??** defines each class operationally.

Table 7: Collapsed three-class role taxonomy. Fine-grained 9-class labels (TRUE-FILL, PARTIAL-FILL, IE, SE, BM, SN, RC, FP, UNCLEAR) are used for qualitative analysis only; all quantitative claims use the 3-class collapse.

Class	Operational criterion
Void-resolution	Paper explicitly bridges A and B under anchor q: proposes new method, mechanism, or synthesis connecting both source directions. Fine-grained: TRUE-FILL or PARTIAL-FILL.
Boundary activity	Paper extends one source direction or provides support, measurement, or naming relevant to the gap without bridging both sides. Fine-grained: IE, SE, BM, or SN.
False positive	Paper is geometrically close but does not address the anchor-conditioned void; topic, references, and framing are off-target. Fine-grained: FP.

Table 8: Role audit (1076 cases, t3+t4+t5; exploratory). VR=void resolution; FP=false positive. 95% Wilson CI in brackets (%).

Source	n	FP	FP CI	VR CI
TVA	204	71%	[65,77]	[1,6]
B1	158	75%	[67,81]	[0,2]
B2	202	67%	[60,73]	[1,6]
NM†	512	84%	[81,87]	[1,4]

†Near-miss (sim$\in [0.75, 0.82)$). CI in %.

An exploratory role audit shows 59–76% false positives across all groups and splits (pilot-scale; n=204 TVA gated cases across t3+t4+t5, role labels from a single LLM). Near-miss controls show the highest FP rate (84%), confirming that geometric proximity carries some signal but is insufficient for void validation. TVA and B1 are indistinguishable; B2 is marginally better. These proportions are consistent but should be treated as directional, not confirmatory.

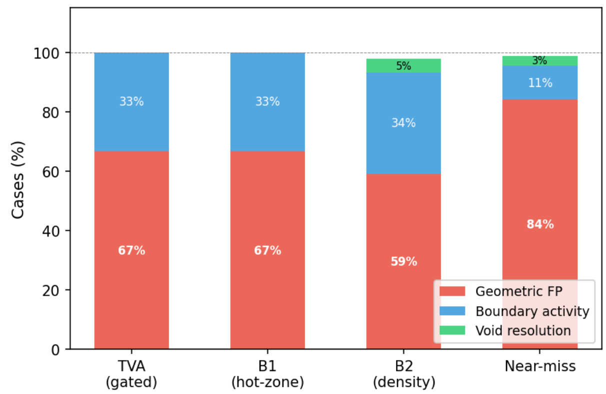

Figure 4: Epistemic role decomposition by source group (t3+t4+t5, n=1076 total). Most raw fills are geometric false positives. Near-miss controls show the highest FP rate.

4.5 Landmark Three-Layer Diagnostic

As a positive-control diagnostic, we select five landmark bridge-type papers (pre-specified, widely cited, 2019–2023) and apply the full three-layer TVV predicate: geometric closure (G), anchor eligibility (E), and LLM-assisted bridge interpretability audit (R). Each landmark uses its publication-year cutoff as the train corpus boundary.

Table 9: Landmark three-layer diagnostic. All five papers clear G and E; none clear R.

Landmark	G	E	R
vLLM / PagedAttention (2023)	**Yes**	**Yes**	No
MLIR (2020)	**Yes**	**Yes**	No
Devign (2019)	**Yes**	**Yes**	No
Ansor (2020)	**Yes**	**Yes**	Weak
FlashAttention (2022)	**Yes**	**Yes**	No

All five landmarks exceed the same-anchor null midpoint p_{95} threshold (G pass) and are anchor-eligible for at least one TVA anchor (E pass). However, LLM bridge audit (R) finds that the corresponding best-void source pairs are semantically incoherent: for example, vLLM's best void pairs a GPU energy survey with a VANET routing paper, not OS paging with LLM serving.

This 5/5 G+E pass, 0/5 R pass pattern is precisely the failure mode TVV is designed to surface. It demonstrates that the three layers are not redundant: geometric proximity and anchor alignment are necessary screening signals but are insufficient without role-aware interpretation. The detector beeps at the right domain, but recovering the correct bridge pair requires the full adjudication pipeline.

4.6 Finding 4: Geometric Impossibility of R-Layer Substitution

We tested whether the LLM-based R layer can be replaced by a closed-form graph-spectral surrogate. Three independent operators were evaluated on the same C1-restricted void population (Table **??**).

Table 10: Three spectral surrogates for the R layer. All fail by distinct mechanisms. NM = non-monotone rate (signal direction wrong). † Degenerate (constant signal): κ not meaningful.

Stage	Operator	N	Δ-signal	NM	κ
A	ΔR global	51k	0	—	$-1.0\dagger$
B	ΔR local	$\sim$328	$9e{-}4$	74%	-0.013
C	$\Delta\kappa_{\mathrm{ORC}}$	$\sim$329	$2.4e{-}3$	43%	0.005
3a-Q1	XEnc anchor	100	—	—	0.10
3a-Q2	XEnc A+B	100	—	—	0.01

Stages B and C additionally share consistent weak negative Pearson correlations with LLM fill labels ($r \approx -0.12$, both $p > 0.20$), suggesting a sub-threshold geometric signal whose signal-to-noise ratio is insufficient for class adjudication under any tested operator.

Stage A: Global k-NN graph (N=51 460). We computed the change in effective resistance $\Delta R / R_{\mathrm{before}}$ between A- and B-side communities before and after inserting candidate filler P into the full BGE-M3 train corpus k-NN graph. $\Delta R/R = 0.0000$ at machine precision across all 100 tested cases. Cohen's $\kappa = -1.0$ vs LLM labels (degenerate: signal is a constant). Root cause: in BGE-M3's anisotropic 1024-dimensional space, the global k-NN graph is nearly fully connected, leaving no detectable resistance contrast.

Stage B: Localized subgraph (N≈328). We restricted the graph to the C1 anchor pool plus k-NN neighborhoods of A, B, and the midpoint (N≈328, well within the regime where von Luxburg's commute-time collapse should not apply). $\Delta R/R$ showed a non-zero but uninformative distribution (mean = 9×10^{-4}, p50 = 6×10^{-4}, p90 = 3×10^{-3}). Critically, 74% of cases showed $\Delta R > 0$ (resistance *increased* after P was inserted), indicating non-monotone behavior: new nodes change the degree distribution of both endpoints, and ΔR can be negative for true bridges and positive for false positives (a network-level Braess-paradox effect). Cohen's $\kappa = -0.013$ vs LLM labels, and Pearson $r(\Delta R, \text{fill}) = -0.13$ ($p = 0.21$). The signal is structurally noise.

Stage C: ORC path curvature (N≈329, n=30). We tested Ollivier-Ricci curvature (ORC) on the same localized subgraph. For each void, we computed mean ORC along the A→B shortest path before and after inserting P, recording $\Delta\kappa_{\text{path}}$. $\Delta\kappa$ showed a distribution with mean 2.4×10^{-3} (p50 = -3.0×10^{-3}, p90 = 2.6×10^{-2}); 43% of cases showed positive $\Delta\kappa$ (analogous to the non-monotone pattern in Stage B). Cohen's $\kappa = 0.005$ vs LLM labels; Pearson $r(\Delta\kappa, \text{fill}) = -0.11$ ($p = 0.57$). ORC path curvature shows no class-discriminative signal.

Stage 3a: Cross-Encoder Reranking. To test whether the impossibility extends beyond bi-encoder geometry, we evaluated BGE-M3's own cross-encoder sibling (`bge-reranker-v2-m3`) on the same 100 cases. The cross-encoder can attend to token-level interactions and thus bypasses the bi-encoder bottleneck. Four query formats were tested: (Q1) anchor text, (Q2) A+B concatenated, (Q3) an explicit bridging prompt, (Q4) min(score(P,A), score(P,B)).

Q1 (anchor) achieves AUC=0.875, confirming the cross-encoder detects anchor relevance (G layer). However, Q2 (joint bridging) achieves AUC=0.352 < 0.5 — *anti-correlated* with fill. Inspecting the 4-class breakdown, INCREMENTAL-EXTENSION and SURVEY-OR-NAMING papers score *higher* on Q2 than PARTIAL-FILL papers (means 0.17/0.19 vs. 0.018), because both are topically related to A or B but only one side. The cross-encoder rewards single-domain proximity, not epistemic bridging.

Cohen's $\kappa = 0.010$ (Q2), 0.102 (Q1), 0.067 (Q3), -0.014 (Q4). No query format reliably discriminates fill from non-fill. Cross-encoder reranking is the 4th independent method to fail.

Synthesis and Impossibility Conjecture. Four independent methods fail across two inference paradigms (geometric operators on Stages A–C, cross-encoder joint attention on Stage 3a) and four failure modes: anisotropy collapse (Stage A), non-monotone sign instability (Stage B), flat curvature (Stage C), and anti-correlated relevance (Stage 3a-Q2). The consistency of failure across inference paths sharing the same BGE-M3 backbone suggests the bridging signal is absent at the representation level, not merely at the bi-encoder readout.

> *Conjecture (Architectural Impossibility).* For retrieval-trained dense encoders under InfoNCE-class objectives, $I(\text{geometric/attention features}; R \mid \phi) \approx 0$ for cross-domain epistemic bridging within C1-restricted anchor pools. This conjecture is grounded in four independent failures with distinct failure modes, ruling out a single-operator explanation. Scope: C2/C3/C4 contribute ≤7% marginal selectivity on random pairs (Diagnostic 1); tighter calibration may alter the void population but is unlikely to rescue spectral surrogates given BGE-M3 anisotropy. Formal upper bounds on I for arbitrary operator

families require adversarial construction and are left to future work.

Implication. The R layer's reliance on text-level semantic assessment is a structural necessity imposed by the architecture of contrastive bi-encoders. The TVV three-layer design (G→E→R) reflects this asymmetry: geometric layers are computationally cheap and architecturally captured by retrieval encoders, while epistemic role requires inference beyond fixed-vector compression.

4.7 Case Studies

The following three cases are selected from the t4/t5 role audit to illustrate why geometric proximity alone does not determine epistemic role. For each case we report the anchor direction, the two source-side communities, why the future paper is geometrically close, and why the role differs.

Case 1 – Partial Fill [?] (2019). Anchor: virtualization and cloud resource management. Source A lies on the VM resource allocation side; Source B lies on vehicular task-scheduling. The future paper studies multi-task offloading over vehicular clouds, combining resource allocation, task scheduling, and cloud/edge execution. Its embedding falls near the midpoint (sim(P, m)=0.835) because it shares vocabulary from both sides. However, its reference list contains 10+ task-offloading citations (B-side) but only one marginal VM reference (A-side). It approaches the bridge but does not fully connect both source communities. Role: PARTIAL-FILL.

Case 2 – Boundary Expansion [?] (2021). Anchor: kernel scheduling and CPU affinity for heterogeneous systems. Source A represents the kernel-scheduler/CPU-affinity side; Source B represents heterogeneous-system optimization. The future paper proposes AI planning heuristics for heterogeneous HPC configuration and is geometrically close to the midpoint (sim(P, m)=0.855) through shared language of scheduling, planning, and resource optimization. Its entire bibliography (44 references) remains within the HPC optimization literature; there are no kernel scheduler, CPU-affinity, or OS-level references. It extends Source B's direction without crossing toward Source A. Role: INCREMENTAL-EXTENSION.

Case 3 – False Positive [?] (2017). Anchor: storage I/O path optimization and NVMe driver design. Despite having the *highest* midpoint similarity of the three cases (sim(P, m)=0.875), the paper studies energy-aware virtual network embedding. The high similarity is driven by generic vocabulary—"virtual", "resource", "energy", "allocation"—shared with cloud infrastructure papers on both source sides, not by substantive connection to storage or NVMe research. Its 19 references contain no storage I/O, NVMe, block-layer, or anchor-domain citations. It is geometrically close but epistemically off-target. Role: FALSE-POSITIVE.

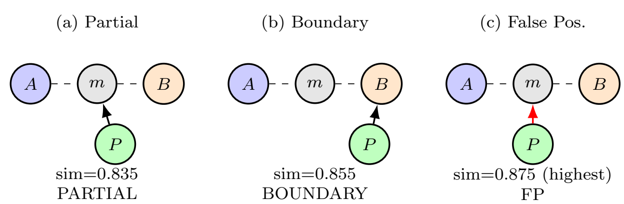

Figure 5: Case-study role patterns. Case (c) has the *highest* sim(P, m) yet is a false positive; Case (a) has lower similarity but partially bridges both source sides. Geometric proximity alone does not determine epistemic role.

5 Discussion

5.1 Raw Fill Fails for Three Reasons

Raw fill conflates void quality with sociotechnical realization rate: (1) hot-zone baselines are biased by local momentum; (2) fixed thresholds are not portable across anchor-density regimes; (3) anchor-eligible future papers are sparse (∼6%), right-censoring many unfilled voids.

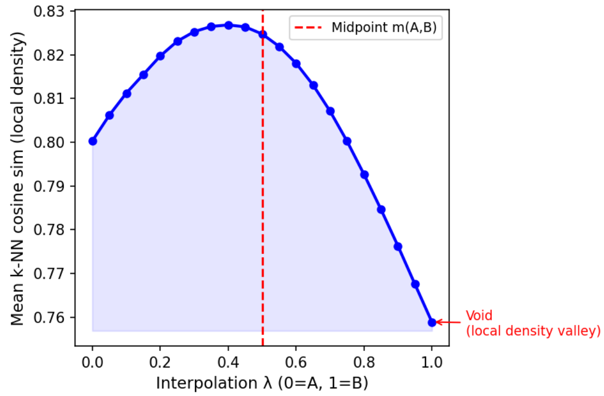

Figure 6: Local density profile along the A–B bridge for the highest-MMR void in anchor sched_opt (t5). The density valley at $\lambda \approx 0.5$ marks the void midpoint region—two dense source communities with a less-occupied bridge.

5.2 B2 Is a Control, Not a Competitor

B2 is a strong counterfactual rather than a weak random baseline: it fixes the anchor and matches local midpoint density, estimating the background rate at which future literature occupies comparable regions without TVA's structured void-selection criteria (C1–C4).

Under calibrated thresholds, TVA reaches parity with B2 but does not outperform it. We interpret this as a conservative result in a low-base-rate, corpus-limited regime: fill events are rare, both methods are exposed to the same small set of future papers, and the current corpus and validation windows may lack the statistical power to distinguish structured void discovery from density-matched background occupancy.

Parity with B2 is in fact the expected outcome under a properly controlled design. B2 matches the anchor and local density of TVA voids, so it estimates the background arrival rate of future papers in comparable regions. A large TVA advantage under this control would imply either strong method-specific predictive signal or residual confounding; we observe neither. Instead, both methods converge to the same low calibrated fill rate, indicating that significant geometric closure is primarily base-rate limited in this corpus.

B2 is intentionally conservative: it makes raw-fill superiority difficult to claim without evidence of signal beyond anchor and density. TVA does not provide that signal at this scale—but it was not designed to win on fill rate. Its value lies in turning a latent region into a named, inspectable, and actionable ex-ante hypothesis (A, B, q, m) before any future paper exists. Raw temporal fill rate is not a sensitive metric for that kind of value.

5.3 Why the R Layer Requires Language Models

Our empirical impossibility result (Finding 4) has a natural theoretical grounding in the form-meaning distinction articulated by [?]: *"A system trained only on form has a priori no way to learn meaning."* Dense retrieval encoders such as BGE-M3 are trained on form-form alignment: they optimize whether two symbol sequences co-occur in relevance-annotated pairs. This is sufficient for topical similarity (G layer, Stage 3a Q1 AUC$=0.875$) but structurally insufficient for adjudicating communicative intent — whether a document's authors synthesize two research traditions rather than merely cite them. This limitation is consistent with work showing that contrastive uniformity forces distinct classes into maximally separated spherical caps [?], and that form-trained dual-encoders fail on compositional benchmarks where surface statistics are insufficient [?].

The key distinction is *compositionality* [?] rather than dimensionality. Bi-encoders compress each paper into a single fixed-length vector, discarding compositional structure (P uses B's method to address A's problem). Cross-encoders add joint attention but still collapse to a scalar readout; the bridging signal is lost at that final compression (Stage 3a Q2 AUC$=0.352$, anti-correlated with fill). Language models preserve the full token sequence and can *dynamically recompose* A- and B-side concepts during inference. The R layer does not require an LLM because bridging is somehow "in a higher dimension" — it requires an LLM because bridging is a relational, compositional property that fixed-vector compression discards.

The Q1 vs Q2 AUC gap on the cross-encoder ($0.875 \rightarrow 0.352$) provides direct empirical evidence of this gap: the encoder can rank by form-level proximity (Q1), but negatively ranks by compositional synthesis (Q2).

5.4 TVA as a Rapid Triage Instrument

The absence of a significant TVA–B2 fill-rate advantage should not be interpreted as an absence of practical utility. B2 is a strong *retrospective control*, not a discovery workflow: it estimates how often density-conditioned regions receive future occupancy. TVA, by contrast, produces a ranked list of inspectable void hypotheses (q, A, B, m) within minutes. Parity with B2 under dependence-aware tests is therefore non-trivial: TVA reaches the background future-occupancy rate of a strong matched control while also providing human-readable source-side structure for follow-up.

TVA is a *detector*, not an oracle. A high TVA signal is analogous to an anomaly reading: it prioritizes candidate locations for inspection, but does not certify epistemic closure. TVV provides the downstream adjudication protocol—calibrated geometric closure, anchor observability, and role-aware assessment—to determine whether a candidate void corresponds to a genuine epistemic gap.

Human-machine division of labor. The combinatorial search for candidate voids is machine-scale: a modest anchor neighborhood of 300 source papers induces 45 000 candidate pairs, and TVA generates 300 ranked void hypotheses across 10 anchors in minutes. Human experts are not designed for exhaustive enumeration at this scale, but are well-suited to adjudicating a compact, ranked list of interpretable (q, A, B, m) candidates. TVA therefore functions as a high-throughput triage step that compresses a machine-scale search space to a human-auditable hypothesis set.

TVA should therefore be evaluated as a *rare-event triage instrument.* Because genuine epistemic closure is a low-base-rate event, most candidate voids will not become validated closures within any finite corpus window. The relevant measure is not absolute conversion rate but *enrichment*: does TVA concentrate observable future-bridging events into a smaller, inspectable candidate set relative to unguided or density-only search? Three rates matter at distinct levels: (i) *detection rate*—does TVA surface the right anchor domain? (ii) *interpretability rate*—does the A/B/q hypothesis make structural sense? and (iii) *realization rate*—does a future paper actually close the predicted gap? Only the first two are directly measurable from pre-publication corpora; realization is what TVV attempts to evaluate, and it is expectedly rare.

Table 11: Division of labor.

Task	Human	TVA/TVV
Enumerate pairs	Infeasible	Automatic
Rank voids	Infeasible	Automatic
Anchor alignment	Difficult	Automatic
Bridge meaning	Strong	Assisted
Technical merit	Strong	Exploratory

Note: TVA produces (q, A, B, m) hypotheses for downstream review; B2 produces matched midpoints to estimate background occupancy. The two serve fundamentally different roles and are not competing discovery systems.

5.5 From Geometric Occupancy to Role Decomposition

Future papers near void midpoints decompose into true fills, boundary expansions, and false positives. The scarcity of true fills reveals that genuine epistemic closure is a much stricter condition than geometric occupancy. *The three layers are operational screens for disentangling three sources of evidence that raw fill rate conflates: geometric occupancy, observability, and epistemic role. They are not claimed to be necessary and sufficient conditions for scientific discovery.*

5.6 Descriptive, Not Evaluative

TVV classifies the type of validation evidence produced by future literature, not the intrinsic merit of papers. Boundary expansion and incremental extension are valuable scientific activities.

5.7 Dynamic Extension (Future Work)

These results suggest a natural extension: instead of asking whether a predicted void is eventually filled within a fixed window, one can classify each newly arriving paper by its pre-insertion topological impact on the existing knowledge space. This dynamic formulation is outside the scope of the present work.

5.8 Threats to Validity

(1) *Embedding dependence*: all metrics use BGE-M3. (2) *Threshold sensitivity*: calibration uses 200 null midpoints per cell. (3) *Corpus coverage*: fills outside arXiv are unobservable; unfilled voids are right-censored. (4) *LLM classification reliability*: role labels are produced by a single LLM (Claude Sonnet 4.5) using a fixed rubric. To assess reliability: (i) blind source-label removal (source group not shown to classifier); (ii) cross-LLM re-annotation on 100 stratified t5 cases (Claude 4.5 vs. GPT-5.2; κ_3=0.44, "moderate"; see §4.3); (iii) a blinded human audit on 60 cases (>80% binary FP agreement, ∼65% 3-class). Epistemic role is inherently interpretive; neither cross-LLM nor human labels constitute ground truth. (5) *Domain specificity*: CS arXiv categories only. (6) *Encoder generality*: the architectural impossibility result is established on BGE-M3 and its cross-encoder sibling. Replication on E5, GTE, or instruction-tuned encoders is left to future work.

6 Related Work

Validation of Predicted Voids. Prior retrospective evaluations in LBD [?, ?, ?] typically validate predicted connections by checking whether future publications instantiate an A–B association. Knowledge-graph completion similarly evaluates whether missing edges are recoverable under a fixed relation schema. These settings differ from void validation: a TVA void is not a predicted edge but an anchor-conditioned geometric vacancy whose future occupancy may represent true bridging, boundary expansion, or false-positive proximity. To our knowledge, prior work has not evaluated embedding-space voids under density-controlled, observability-aware, and role-aware temporal validation. TVV addresses this gap.

Literature-Based Discovery. Swanson's ABC model [?] and subsequent LBD work [?, ?] validate predictions by checking future publication appearance. TVV extends this tradition by requiring epistemic bridging role, not merely geometric proximity.

Novelty and First-Story Detection. Topic Detection and Tracking [?] introduced first-story detection; novelty detection in IR [?] measures whether a document adds new information relative to a query set. Our first-filler intuition shares the event-ordering structure but operates on knowledge structure rather than event reporting.

Structural Holes and Scientometrics. Burt's structural hole theory [?] shows brokers spanning gaps accumulate informational advantage. Co-citation analysis [?] and our density-matched baseline directly control for hot-zone momentum.

Dense-Region Bias in Embeddings. Anisotropy in high-dimensional text embeddings [?] compresses cosine similarities into a narrow range. We show this anisotropy makes fixed thresholds non-portable in downstream void validation. Our use of "topological" is operational rather than homological [?, ?].

Scientific Claim Verification. Epistemic classification of scientific contributions—argumentative zoning [?], overclaiming detection [?]—provides precedents for our role-aware rubric. [?] introduced scientific fact verification as a retrieval task; our R layer extends this from claim-level verification (does evidence support a claim?) to bridge-level role assessment (does P bridge A and B under anchor q?). The form-meaning gap articulated by [?] provides the theoretical backdrop for our architectural impossibility result (§??).

7 Conclusion

Temporal validation of predicted research voids requires separating geometric occupancy, density bias, anchor observability, and epistemic role. Under density matching, TVA's apparent disadvantage disappears but is not statistically significant. Under null-calibrated thresholds, fill rates collapse to ∼0.7%—most raw fills were null-zone noise. LLM-assisted role-aware classification suggests that genuine epistemic bridging is rare even among geometrically close, anchor-eligible future papers.

We propose a three-layer validation framework and leave two open questions for future work: (1) scalable role-aware validation without prohibitive LLM cost; (2) dynamic event-driven extensions in which each arriving paper updates knowledge-space state.

A Reproducibility Protocol

Corpus. arXiv metadata (title, abstract, categories, update_date) for categories cs.OS, cs.AR, cs.PL, cs.SE, cs.DC, cs.NI, cs.CR, cs.AI, cs.LG. Filtered by category; year assigned from update_date.

Temporal splits. Three non-overlapping splits. For split s with cutoff T_s: train = papers with year $< T_s$; val = papers with year $\in [T_s, T_s + 5)$.

Embedding. BGE-M3 (BAAI/bge-m3, 1024-dimensional, cosine similarity, offline/local mode). Papers embedded as "*title abstract*" concatenation.

TVA void generation. For each anchor q in Table **??** and each split: (1) embed anchor text with BGE-M3; (2) retrieve top-300 cosine-nearest train papers (C1 pool); (3) enumerate all pairs and apply C1–C4 conditions; (4) rank by MMR score and output top-30 voids. No anchor added or removed after validation runs began.

Table 12: TVA anchor list (pre-specified, fixed across all splits).

Key	Anchor phrase
sched_opt	kernel scheduler optimization and CPU affinity for multicore systems
mm_vm	virtual memory management and page reclamation in operating systems
ebpf_obs	eBPF programs for kernel tracing and system call observability
hwsw_x86	hardware-software co-design for x86 processor microarchitecture
net_io	high-performance network I/O and packet processing in the kernel
sync_lock	synchronization primitives and lock-free data structures
mem_alloc	memory allocator design and heap fragmentation in systems software
virt_hyp	virtualization and hypervisor design for cloud workloads
power_mgmt	dynamic power management and CPU frequency scaling
storage_io	storage I/O path optimization and NVMe device drivers

B1 baseline. For each anchor, sample 20 random pairs without replacement from the C1 pool (seed 42). No void conditions applied.

B2 density-matched baseline. For each TVA void V, compute midpoint density $\rho(m_V)$ (mean cosine similarity to k=20 nearest train neighbors). Sample 300 B1-style pairs from the same anchor's C1 pool (seed 99). Select the pair whose midpoint density is closest to $\rho(m_V)$ as the B2 counterpart.

Local density.

$$\rho(m) = \frac{1}{k}\sum_{i=1}^{k} \mathrm{sim}(m, \mathrm{kNN}_i(m)), \quad k{=}20.$$

Anchor-eligibility threshold. $\tau_q = \max(Q_{80}^{\mathrm{val}}, Q_{90}^{\mathrm{train}})$, computed per anchor and split.

Null-calibrated threshold. For each cell (anchor q, density bucket $b \in \{\text{low, mid, high}\}$, split t): (1) generate n=1000 null midpoints from C1 pool; (2) split 50/50 calibration/held-out; (3) compute $Z(m_0) = \max_{P\in\mathrm{Val}_q} \mathrm{sim}(P, m_0)$ for each null; (4) $\tau_{\mathrm{fill}} = s_{(k)}$ where $k = \lceil (n_{\mathrm{cal}}+1)(1-\alpha)\rceil$ (finite-sample conformal quantile), α=0.05.

Statistical tests. Primary inference uses *hierarchical cluster bootstrap* (outer: resample anchors with replacement; inner: resample voids within each anchor) for TVA–B2 difference in percentage points. Secondary check: *anchor-level sign-flip permutation test* (randomly flip sign of per-anchor mean TVA–B2 difference; n_{perm}=10000). Both report two-sided p-values.

Role annotation. Model: Claude Sonnet 4.5, temperature 0.1. Input: anchor text, source A title+abstract (truncated), source B title+abstract, candidate filler title+abstract. Output: JSON with role label, confidence, and six ordinal scores. Source group labels are not shown (blind classification). Final labels use the collapsed three-class taxonomy (Table ??). Prompt and full rubric text are available upon request.

Code and data availability. Experiment scripts, prompts, and configuration files are maintained in an internal repository at Intel Corporation. They will be made available upon publication subject to internal review and export control clearance. Derived labels for the role-classified 1,076 cases will be released with the camera-ready version.